Data Processing & QC

Unlike a Seaglider, whose basestation QC's every dive automatically (see Seaglider Data Quality Control), a Slocum has **no built-in data QC**. The glider records raw sensor values in compact binary files. The SFMC (or older Dockserver) decodes and plots them for piloting. Everything else is up to the user: converting the files, merging flight and science data, quality control and corrections. Teledyne itself describes the SFMC as a piloting and viewing tool, not the place for full-mission analysis.

The good news is that the community has built a solid open-source toolchain for this. This page walks through the chain from the glider's files to QC'd data.

Source

Paraphrased from the Teledyne Webb Research *Slocum G3 Glider Operators Manual* (science data logging and data retrieval chapters), the Slocum `masterdata`, Teledyne's *Processing Binary Glider Data Outside of the SFMC* notes and file-compression notes, the Webb Research glider forum (data processing, cache-file and merging threads), the UG2 Slack #data, general and #piloting channels (2022–2026), and operator correspondence. Tool behaviour changes between releases: check each tool's documentation, and defer to Teledyne for the glider side.

The data flow at a glance

```
flowchart LR
    G["Glider<br/>flight: dbd sbd mbd mlg<br/>science: ebd tbd nbd nlg"]
-->|"Iridium (sbd/tbd)<br/>or FreeWave / cable (all)"| S["SFMC /
Dockserver<br/>(decode + plots for piloting)"]
    S --> D["Decompress<br/>(compressed files)"]
    D --> C["Convert binary → ASCII / NetCDF<br/>(needs cache files)"]
    C --> M["Merge flight + science<br/>on time"]
    M --> Q["QC + corrections<br/>(user / toolbox)"]
    Q --> O["Users, data centres<br/>(GDAC, Coriolis, ERDDAP…)"]
```